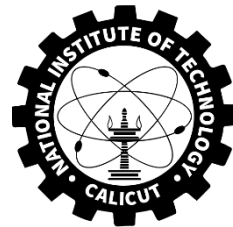


Quality, Reliability & Maintenance ME3327E



तमसो मा ज्योतिर्गमय

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Feature	Discrete Probability Distribution	Continuous Probability Distribution
Definition	Describes the probability of outcomes that can take on countable distinct values.	Describes the probability of outcomes that can take on any value within a continuous range or interval .
Possible Values	Finite or countably infinite (e.g., 0, 1, 2, 3...)	Uncountably infinite (e.g., any real number between a and b)
Mathematical Function	Probability Mass Function (PMF) : Gives the exact probability of a specific value, $P(X = x)$.	Probability Density Function (PDF) : Gives the <i>density</i> at a point; probabilities are found by calculating the area under the curve.
Probability of an Exact Value	$P(X = x) \geq 0$ (Can be a positive number)	Always zero ($P(X = x) = 0$ because there are infinitely many points). We only measure probability over intervals, $P(a \leq X \leq b)$.
Cumulative Function	Cumulative Distribution Function (CDF) : Step function that sums discrete probabilities.	Cumulative Distribution Function (CDF) : Continuous, smooth curve representing the integral of the PDF.
Common Examples	Binomial, Poisson, Hypergeometric, Geometric, Bernoulli	Normal (Gaussian), Uniform, Exponential, Student's t, Chi-Square

Feature	Hypergeometric Distribution	Binomial Distribution	Poisson Distribution
Type	Discrete	Discrete	Discrete
Sampling Method	Without replacement (trials are dependent)	With replacement (trials are independent)	Counts events occurring continuously in time/space
Number of Trials	Fixed, finite population (N)	Fixed number of independent trials (n)	No fixed number of trials; focuses on rate (λ)
Probability of Success	Changes with each draw (p is not constant)	Constant (p) for every trial	Average rate (λ) of occurrence is constant
Key Parameters	N (population size), K (successes in population), n (sample size)	n (number of trials), p (probability of success)	λ (average rate or expected number of events)

Continuous Distributions

Continuous random variables may assume an infinite number of values over a finite or infinite range. The probability distribution of a continuous random variable X is often called the *probability density function* $f(x)$. The total area under the probability density function is 1.

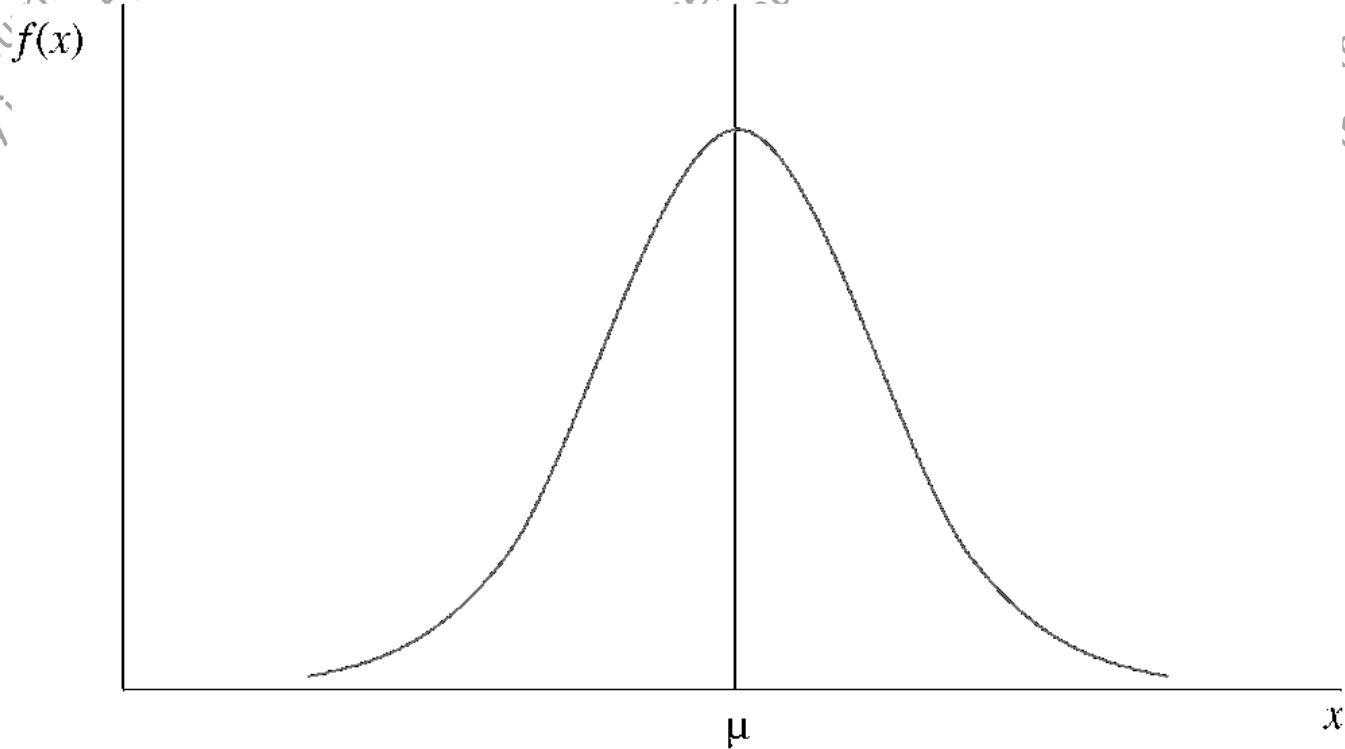
Normal Distribution

The most widely used distribution in the theory of statistical quality control is the *normal distribution*. The probability density function of a normal random variable is given by

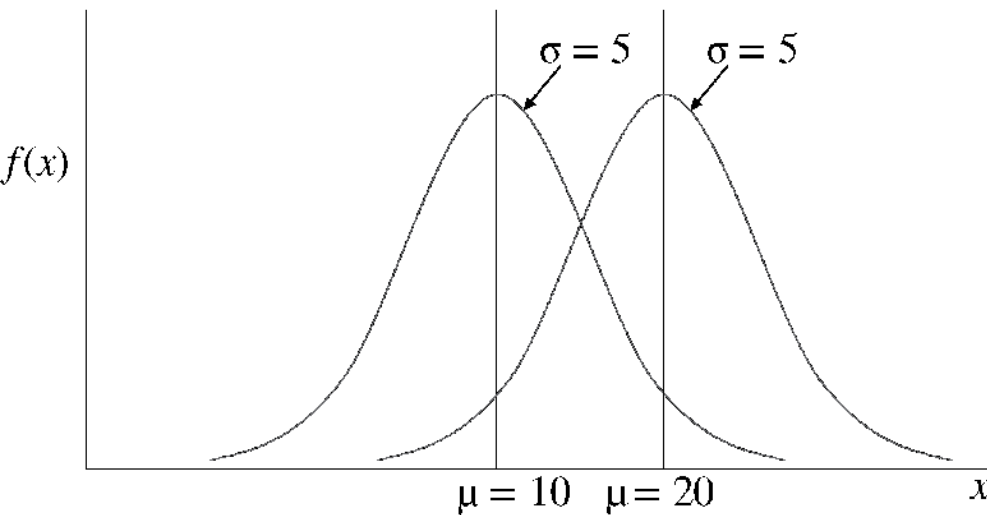
$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left[\frac{-(x - \mu)^2}{2\sigma^2} \right], -\infty < x < \infty$$

Where μ is the population mean and σ is the population standard deviation.

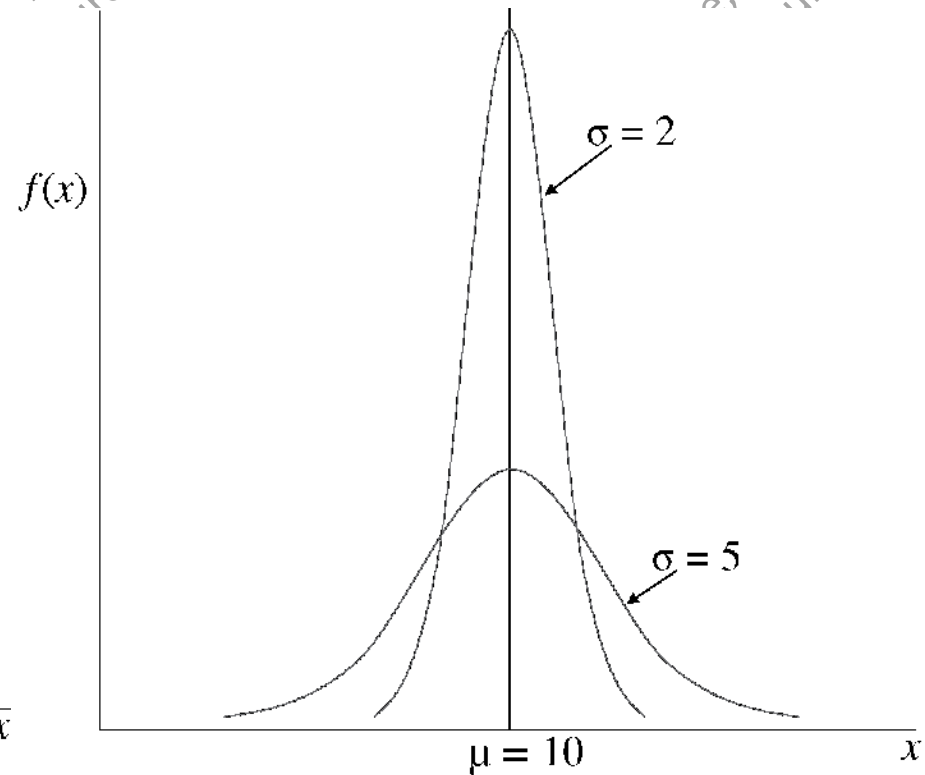
Normal Distribution



Normal Distribution



(a) Effect of a change in the mean



(b) Effect of a change in the standard deviation

Effects of the parameters μ and σ^2 on the normal distribution.

Normal Distribution

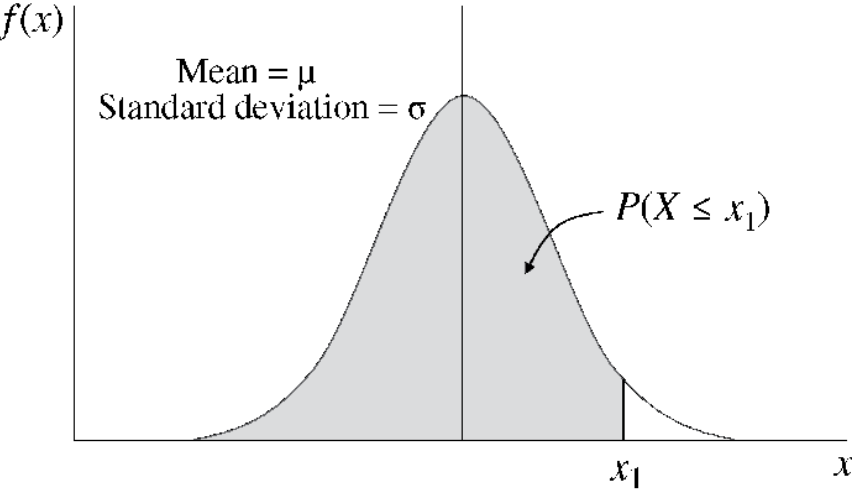
The standardized normal random variable Z is given by,

$$Z = \frac{X - \mu}{\sigma}$$

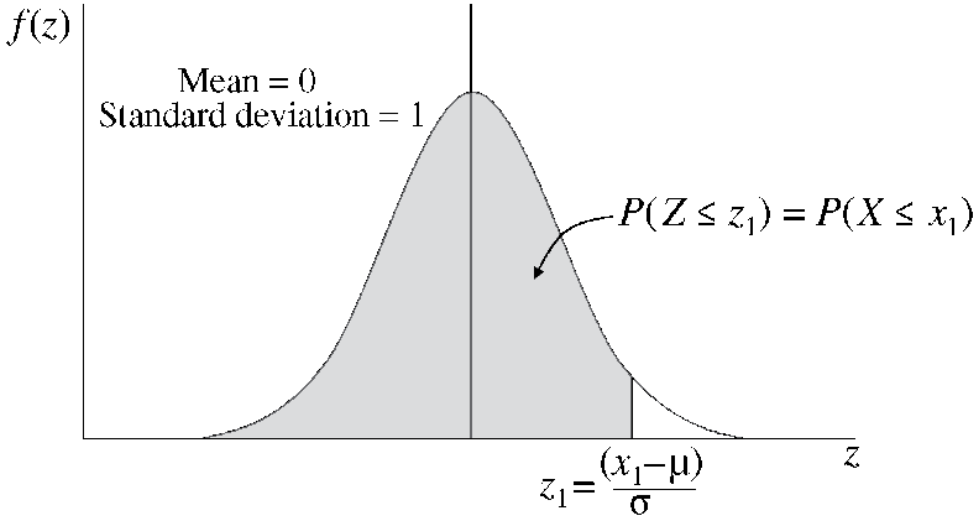
Density function is given by, $f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, -\infty < z < \infty$

The cumulative distribution function of Z is,

$$\Phi(z) = F(z) = \int_{-\infty}^z f(t)dt$$



(a) General form



(b) Standardized form

Exercise



The length of a machined part is known to have a normal distribution with a mean of 100 mm and a standard deviation of 2 mm.

- a) What proportion of the parts will be above 103.3 mm?
- b) What proportion of the output will be between 98.5 and 102.0 mm?
- c) What proportion of the parts will be shorter than 96.5 mm?
- d) It is important that not many of the parts exceed the desired length. If a manager stipulates that no more than 5% of the parts should be oversized, what specification limit should be recommended?

Exponential Distribution

The *exponential distribution* is used in reliability analysis to describe the time to the failure of a component or system. Its *probability density function* is given by $f(x) = \lambda e^{-\lambda x}, x \geq 0$

The mean and the variance of an exponential random variable are given by $\mu = \frac{1}{\lambda}, \sigma^2 = \frac{1}{\lambda^2}$

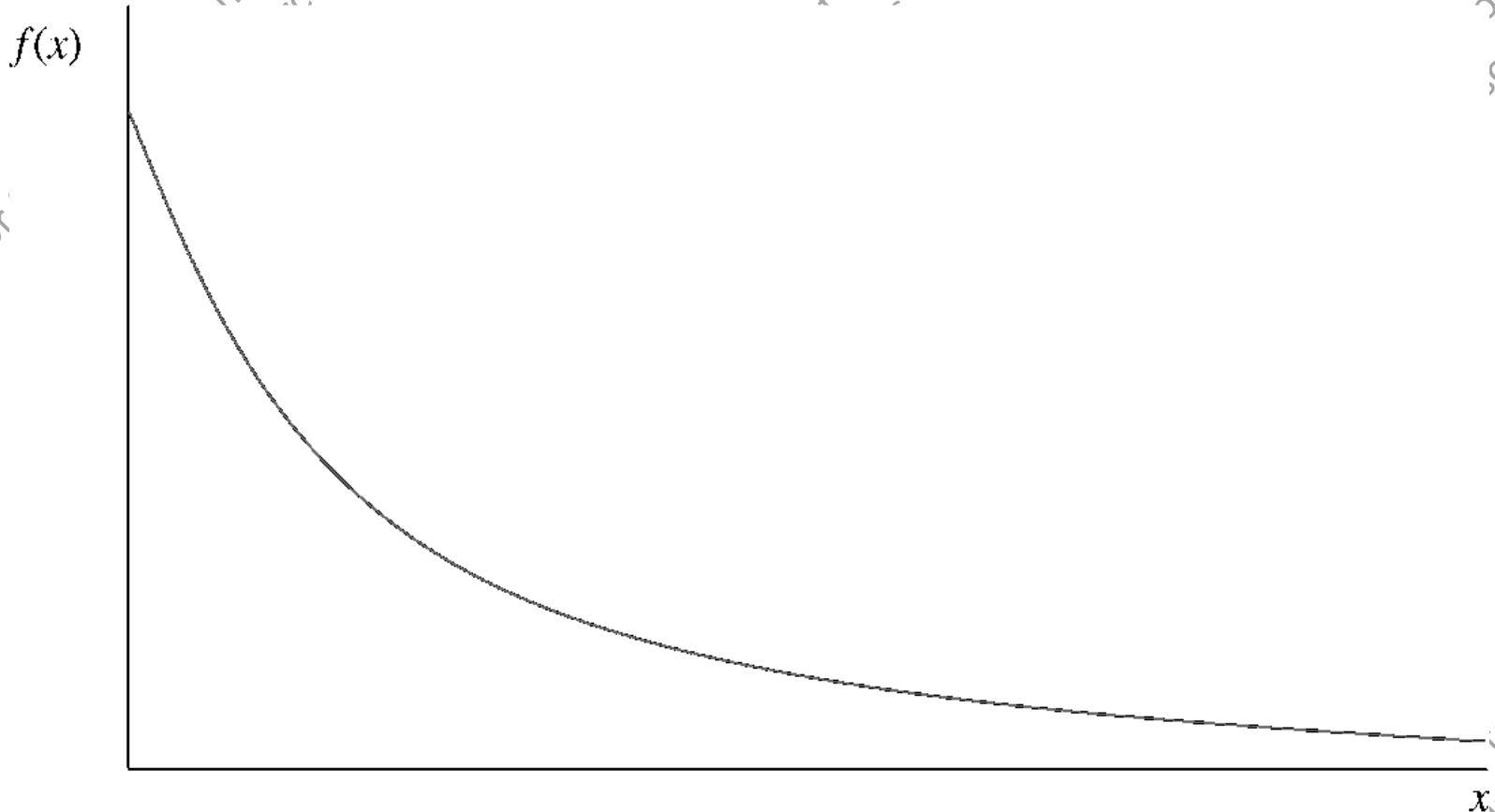
The exponential cumulative distribution function is obtained as follows:

$$F(x) = P(X \leq x) = \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}$$

An *exponential distribution* has the property of being memoryless. This means that the probability of a component's exceeding $s + t$ time units, given that it has lasted t time units, is the same as the probability of the life exceeding s time units. Mathematically, this property may be represented as

$$P(X > s + t | X > t) = P(X > s), \text{ for all } s \text{ and } t \geq 0$$

Exponential Distribution



Exponential density function

Exercise

It is known that a battery for a video game has an average life of 500 hours (h). The failures of batteries are known to be random and independent and may be described by an exponential distribution.

- Find the probability that a battery will last at least 600 hours.
- Find the probability of a battery failing within 200 hours.
- Find the probability of a battery lasting between 300 and 600 hours.
- Find the standard deviation of the life of a battery.
- If it is known that a battery has lasted 300 hours, what is the probability that it will last at least 500 hours?